

Task 4

$$J(\hat{y}) = -\frac{1}{m} \sum_{i=1}^m y_i \log(y_i) + (1-y_i) \log(1-y_i)$$

by chain rule

$$\frac{\partial J}{\partial w_j} = \frac{\partial J}{\partial h} \cdot \frac{\partial h}{\partial z} \cdot \frac{\partial z}{\partial w_j}$$

$$h = \frac{y}{1+e^{-z}} \rightarrow \frac{\partial h}{\partial z} = h(1-h)$$

$$z = w_0 x_0 + w_1 x_1 + w_2 x_2 \quad \frac{\partial z}{\partial w_j} = x_j$$

$$\frac{\partial J}{\partial h} = \left[\frac{-y}{h} - \left(-\frac{1-y}{1-h} \right) \right]$$

$$= \frac{-y}{h} + \frac{1-y}{1-h}$$

$$\left[\frac{-y}{h} + \frac{1-y}{1-h} \right] \times (h + (1-h)) \times x_j$$

$$(-y(h) + (1-y)h) \times x_j$$

$$(-y + yh + h - yh) \times x_j$$

$$= (h - y) \times x_j$$

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